DISTRIBUTION :

Objectives :

(1) Reliable power supply to all the consumers.

(2) Uniform vtg to all the consumers.

Classification :

(1) DC Distribution - used in S/S and Generating S/S for equipment protection.

(2) AC Distribution - used in public utility.

Methods of source :

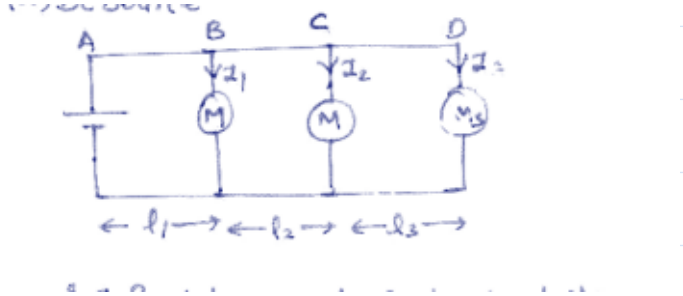
(1) Source fed from one end - Radial system

(2) Source fed from both ends

(3) Ring system

(4) Uniform loading.

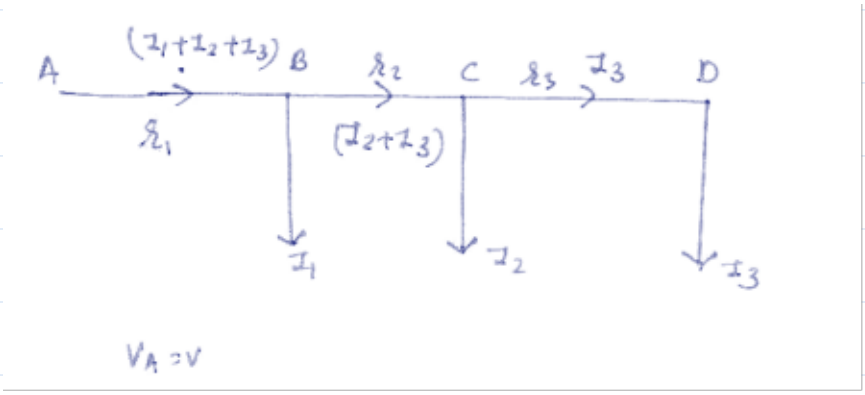
(1) Radial system : (a) DC source



DC source feeding A-B-C-D with loads I1 (motor), I2 (motor), I3; section lengths l1, l2, l3

r = Resistance of each conductor Ω/m ; 2r = Loop (or) Distributor Resistance Ω/m

$R_1 = (2rl_1)$, $R_2 = (2rl_2)$, $R_3 = (2rl_3)$



Distributor A-B-C-D: section currents (I1+I2+I3), (I2+I3), I3 through R1, R2, R3; tap-off loads I1, I2, I3

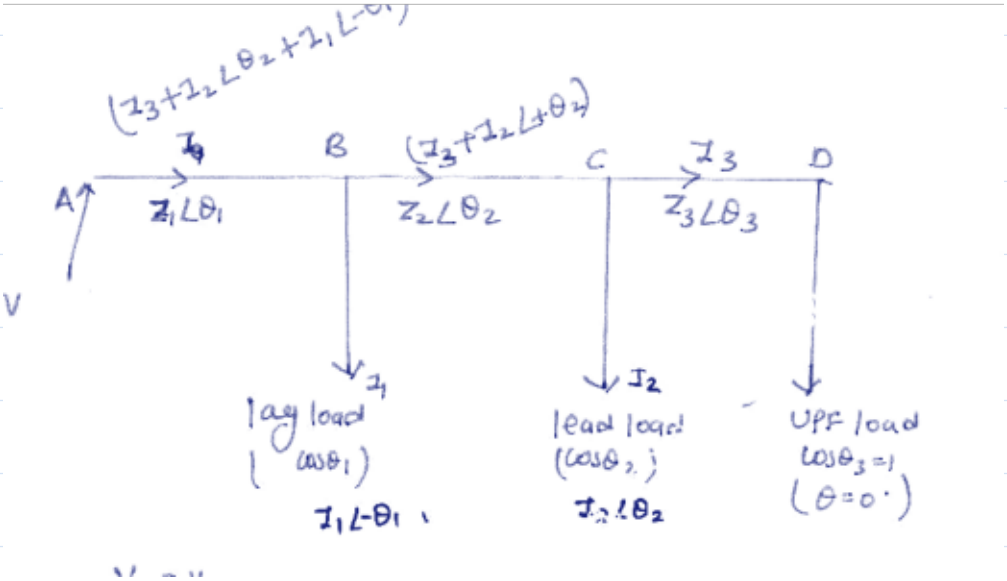
$$V_A = V\ ;\ V_B = V - (I_1 + I_2 + I_3)R_1$$

$$V_C = V_B - (I_2 + I_3)R_2$$

$$V_D = V_C - I_3 R_3$$

$$V_{AD} = V_A - V_D = \text{Vtg drop of distributor.}$$

(b) AC source :



Distributor A-B-C-D with impedances Z1 angle th1, Z2 angle th2, Z3 angle th3; loads I1 lag (cos th1), I2 lead (cos th2), I3 UPF (cos th3 = 1)

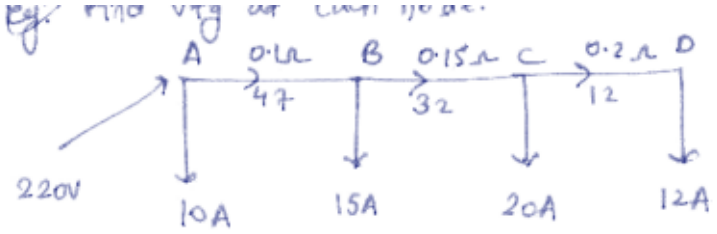
$$V_A = V$$

$$V_B = V - (I_3 + I_2\angle\theta_2 + I_1\angle -\theta_1) Z_1\angle\theta_1$$

$$V_C = V_B - (I_3 + I_2\angle\theta_2) Z_2\angle\theta_2$$

$$V_D = V_C - (I_3 Z_3\angle\theta_3) \; ; \; V_{AD} = V_A - V_D$$

Eg: Find vtg at each node.



220 V at A; sections A-B 0.1 ohm (47 A), B-C 0.15 ohm (32 A), C-D 0.2 ohm (12 A); loads 10 A, 15 A, 20 A, 12 A

$V_A = 220\ V$

$V_B = 220 - 47 \times 0.1 = 215.3\ V$

$V_C = 215.3 - 32 \times 0.15 = 210.5\ V$

$V_D = 210.5 - 12 \times 0.2 = 208.1\ V$

$V_{AD} = 220 - 208.1 = 11.9\ V$

Advantages :

(1) No. of lines are less so power loss is less.

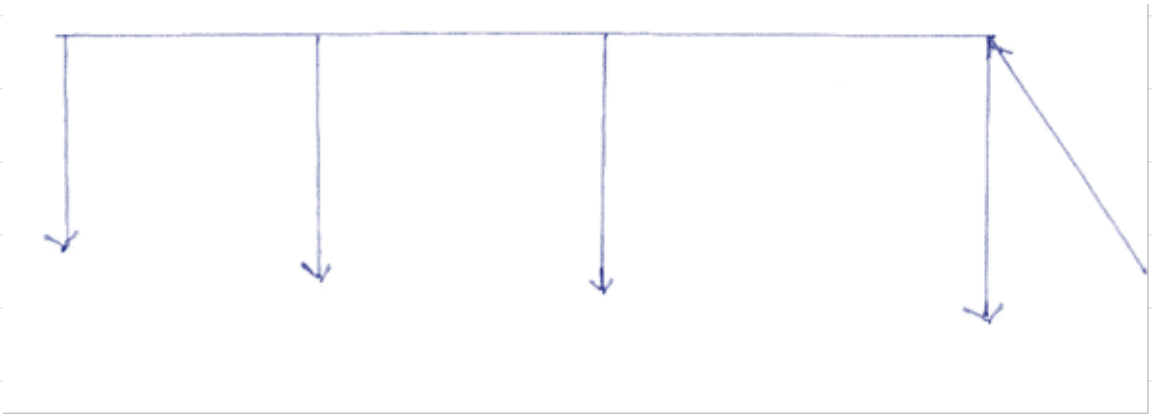
(2) In AC system used as load frequency control.

Disadvantages :

(1) Not reliable

(2) Not uniform vtg.

Source fed from both ends :



Distributor with loads and sources at both ends

Procedure for calculation of Node of minimum potential :

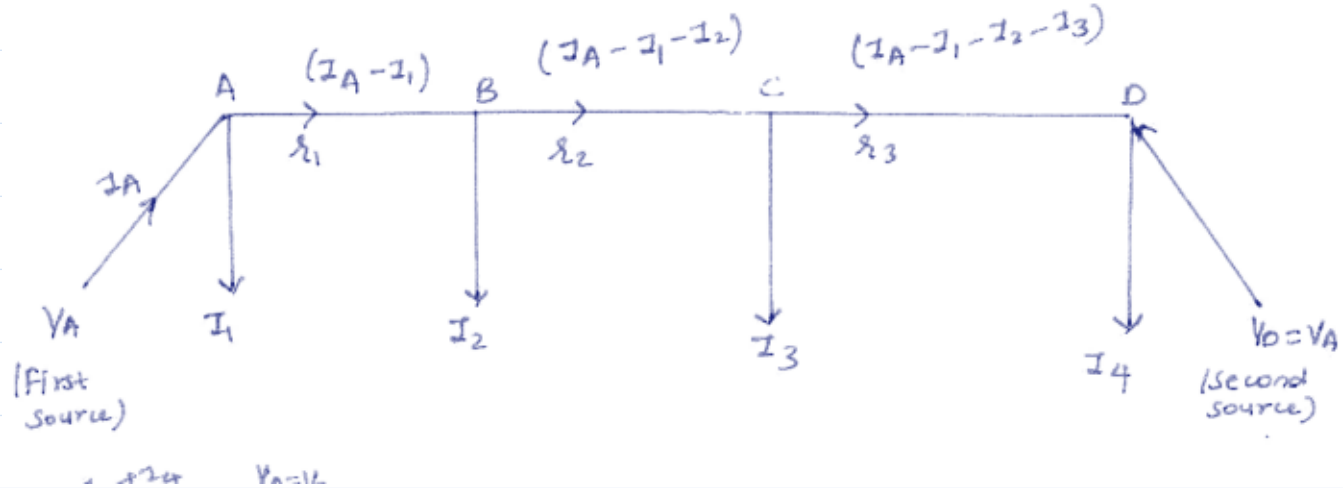
(1) Assume I_A from V_A

(2) Calculate I_A using $(V_A - V_D) = (I_A - I_1)r_1 + (I_A - I_1 - I_2)r_2 + (I_A - I_1 - I_2 - I_3)r_3$

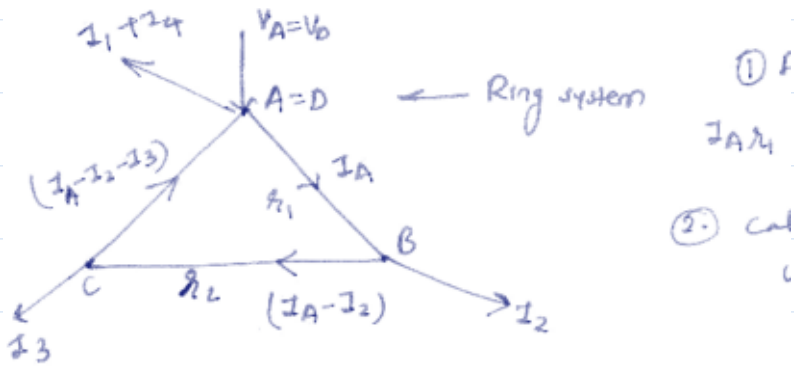
(3) Substitute I_A in $(I_A - I_1 - I_2 - I_3)$, $(I_A - I_1 - I_2)$, $(I_A - I_1)$ and check for first change sign.

(4) Node of 1st sign change = Node of minimum potential.

(5) Apply KVL to calculate vtg at each node.



Distributor A-B-C-D with loads I1..I4: first source VA feeds IA, second source VD = VA; section currents (IA-I1), (IA-I1-I2), (IA-I1-I2-I3)



Ring system A = D, B, C with section resistances r1, r2 and currents IA, (IA-I2), (IA-I2-I3); loads I1 + I4, I2, I3

Ring system : (1) find I_A : $I_A r_1 + (I_A - I_2) r_2 + (I_A - I_2 - I_3) r_3 = 0$; (2) calculate vtg at each node using KVL

Advantage :

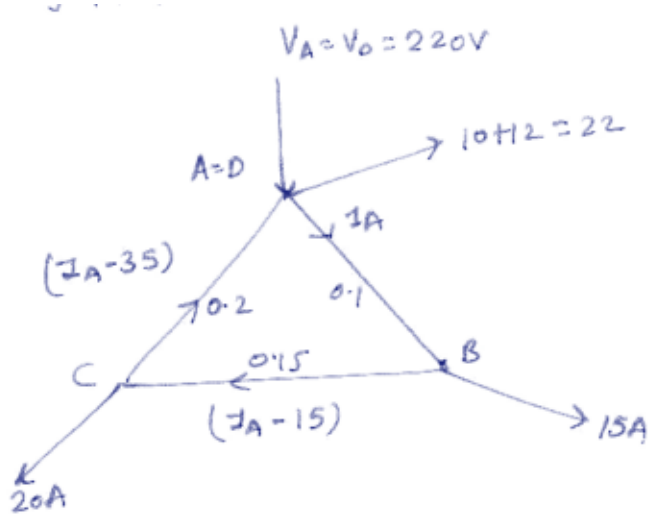
(1) Better vtg profile compared to radial system

(2) Reliable

Disadvantage :

(1) One of the consumers experiences low vtg

Ring system



$V_A = 220 = V_D$

$V_B = 220 - 20.55 \times 0.1 = 217.94$ V

$V_C = 217.94 - (20.55-15) \times 0.15$

$= 217.11$ V

$I_A(0.1) + (I_A-15)0.15 + (I_A-35)0.2 = 0$

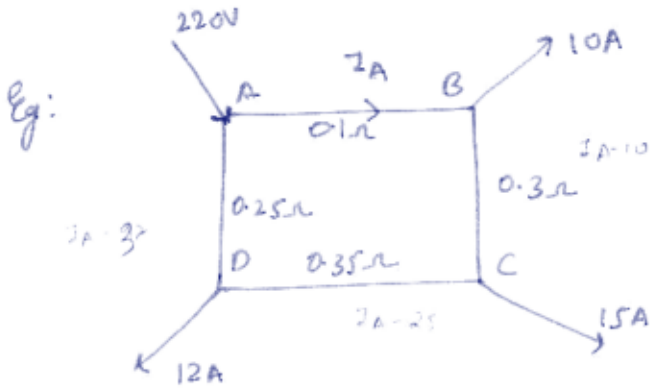
Ring A = D, B, C fed at 220 V; A-B 0.1 ohm (IA), B-C 0.15 ohm (IA-15), C-A 0.2 ohm (IA-35); loads 10+12 = 22 A at A/D, 15 A at B, 20 A at C

$$I_A(0.1) + (I_A - 15)0.15 + (I_A - 35)0.2 = 0$$

$$I_A = \frac{7 + 15 \times 0.15}{0.1 + 0.15 + 0.2} = 20.55 \text{ A}$$

$$V_B = 220 - 20.55 \times 0.1 = 217.94 \text{ V}; \; V_C = 217.94 - (20.55 - 15) \times 0.15 = 217.11 \text{ V}$$

Eg:



$+ I_A(0.1) + (I_A-10)0.3 + (I_A-25)0.35$

$+ (I_A-37)0.25 = 0$

$I_A\,(0.1+1+0.3+0.35+0.25)$

$= 10 \times 0.3 + 25 \times 0.35 + 37 \times 0.25$

$I_A = 21$

$V_B = 220 - 0.1 \times 21 = 217.9$

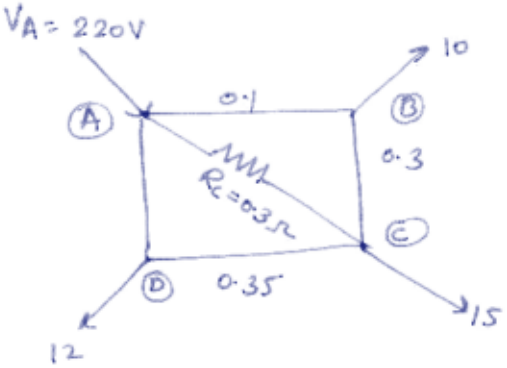
Square ring A-B-C-D at 220 V: A-B 0.1 ohm, B-C 0.3 ohm, C-D 0.35 ohm, D-A 0.25 ohm; loads 10 A at B, 15 A at C, 12 A at D

$$I_A(0.1) + (I_A - 10)0.3 + (I_A - 25)0.35 + (I_A - 37)0.25 = 0$$

$$I_A(0.1 + 0.3 + 0.35 + 0.25) = 10 \times 0.3 + 25 \times 0.35 + 37 \times 0.25 \rightarrow I_A = 21 \text{ A}$$

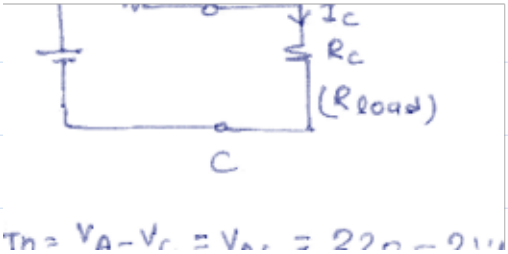
$V_B = 220 - 0.1 \times 21 = 217.9$; $V_C = 217.9 - 0.3 \times 11 = 214.6$; $V_D = 214.6 + 4 \times 0.35 = 216$

Ring system with Interconnector [R_c]



Ring A-B-C-D at 220 V with interconnector Rc = 0.3 ohm between A and C; loads 10 (B), 15 (C), 12 (D)

NOTE: R_c value must be less than least of two parallel paths, i.e. $R_c < 0.4\Omega$ [0.1 + 0.3 = 0.4]. Assume $R_c = 0.3\,\Omega$

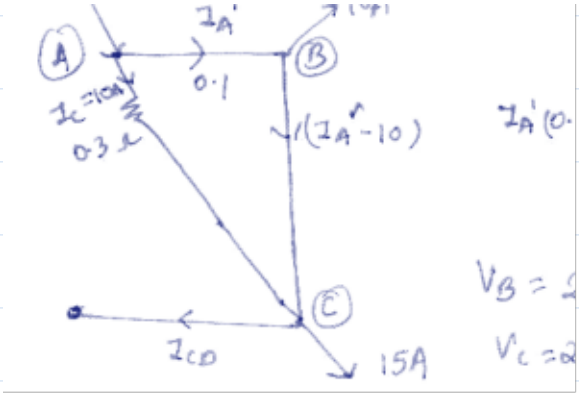


Thevenin equivalent seen from A and C: VTh, RTh feeding interconnector Rc (load)

$$V_{Th} = V_A - V_C = V_{AC} = 220 - 214.6 = 5.4\ V$$

$$R_{Th} = 0.4 \,//\, 0.6 = 0.24\,\Omega$$

$$I_c = \frac{V_{Th}}{R_{Th} + R_c} = \frac{5.4}{0.24 + 0.3} = 10\ A$$



Ring with interconnector carrying Ic = 10 A from A to C; I_A' in A-B 0.1 ohm, (I_A' - 10) in B-C; load 10 A at B, 15 A at C; ICD = 0

$$I_{CD} = 0 ; I_A(0.1) + (I_A - 10)0.3 = 10 \times 0.3 \rightarrow I_A = 15\ A$$

$$V_B = 220 - 15 \times 0.1 = 218.5\ V ; V_C = 218.5 - (15 - 10) \times 0.3 = 217\ V ;$$

$$V_D = V_A - I_{CD} \times 0.35 = 217$$

Advantage :

(1) Uniform vtg is maintained to many of the consumers

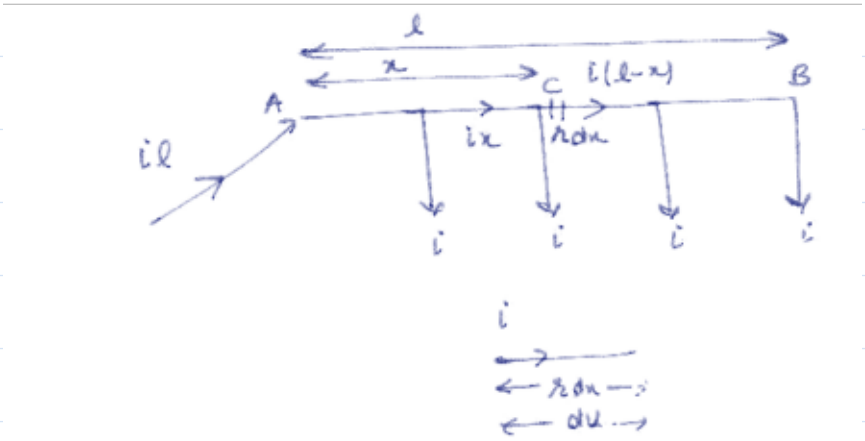
(2) Reliable

Disadvantage :

No. of lines are more so power loss is more.

Uniform loading

(1) Source fed from one end.

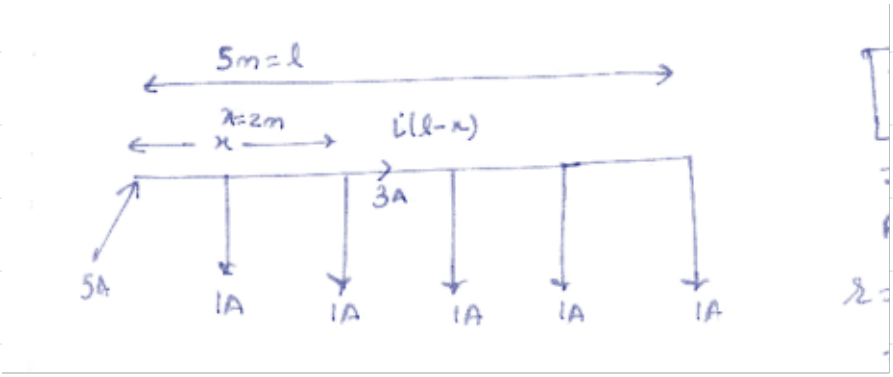


Distributor A-B of length l with uniform load i per unit length; point C at distance x from A; current i(l - x) beyond C; element dx of resistance r dx

$$dV = i(l-x)r\,dx; \qquad V_{AB} = \int_0^l i(l-x)r\,dx = \frac{1}{2}(il)(rl)$$

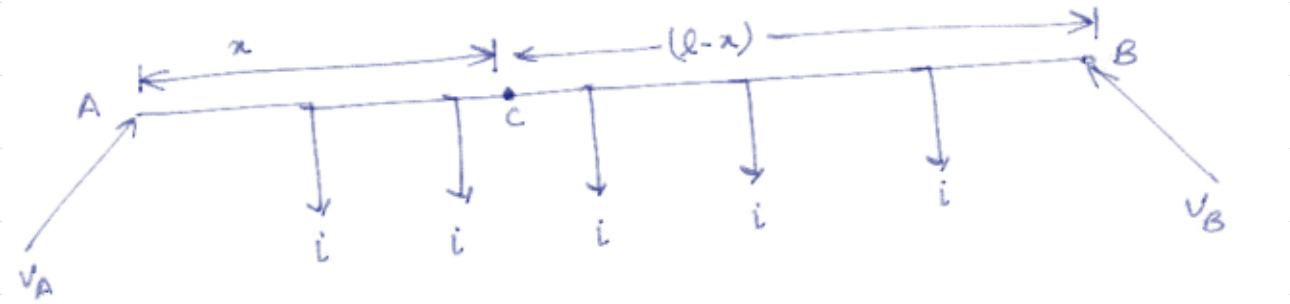
$$\boxed{V_{AB} = \frac{1}{2}IR}$$

$I = i \times l$; $R = l \times r$; r = loop resistance = 2 x per conductor resistance



Numerical illustration: l = 5 m, x = 2 m, 5 A fed at A, 1 A tapped per metre, 3 A flowing at x = 2 m

Distance of minimum potential node for uniformly loaded distributor fed from both ends :



Distributor A-B with uniform load i per unit length, VA at A and VB at B; minimum point C at distance x from A and (l - x) from B

* Assume min vtg at C at a distance x from A. Current supplied by A to C = ix
* Vtg drop in section AC = $V_{AC} = \frac{irx^2}{2}$; length of distributor = l ; current supplied by B to C = i(l - x)
* $V_{BC} = \frac{ir(l-x)^2}{2}$
* $V_{AC} = V_A - V_C \rightarrow V_C = V_A - V_{AC} = V_A - \frac{irx^2}{2}$ -(1)

$$* V_{BC} = V_B - V_C \rightarrow V_C = V_B - V_{BC} = V_B - \frac{ir(l-x)^2}{2} \quad -(2)$$

$$\boxed{x = \frac{V_A - V_B}{irl} + \frac{l}{2}} \quad = \text{point of minimum potential.}$$

* r = Distributor resistance = $2r_1$; r_1 = Resistance of each conductor Ω/m

Eg: LED street lights between Kalusarai and IIT is of 2-wire dc distributor of 500 m long fed from both ends and is loaded uniformly at the rate of 1 A/m. At Kalusarai is maintained at 255 V and IIT 250 V. Resistance is 0.1 Ω/km/conductor.

(1) Find point of min vtg and its value.

(2) The currents supplied from Kalusarai and IIT.

$$r_1 = \frac{0.1}{1000} = 0.0001\ \Omega/m/\text{conductor} \; ; \; r = 2r_1 = 0.0002$$

$$x = \frac{V_A - V_B}{irl} + \frac{l}{2} = \frac{255 - 250}{1 \times 0.0002 \times 500} + \frac{500}{2} = 300\ m$$

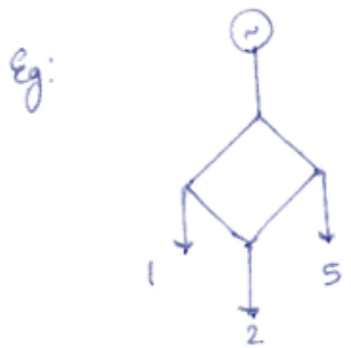
x = min. node vtg is C from A

$$V_C = V_A - \frac{irx^2}{2} = 255 - \frac{1 \times 0.0002 \times (300)^2}{2} = 246\ V$$

Current supplied by (Node A - Kalusarai) = ix = 1 x 300 = 300 A

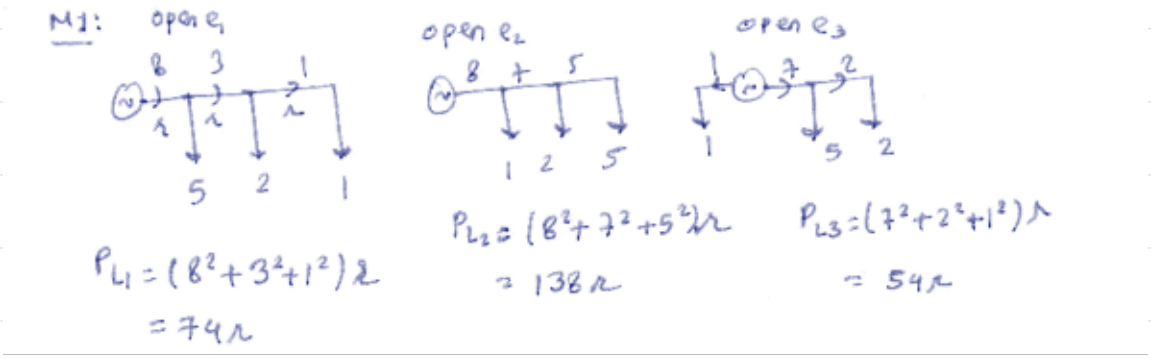
Current supplied by (Node B - IIT) = i(l - x) = 1 x 200 = 200 A

Eg:



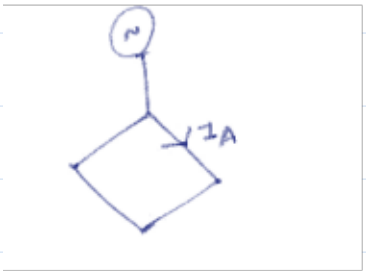
Source feeding a closed loop with loads 1, 2 and 5 at three nodes

M1 : open e_1, e_2, e_3 in turn and find the loss



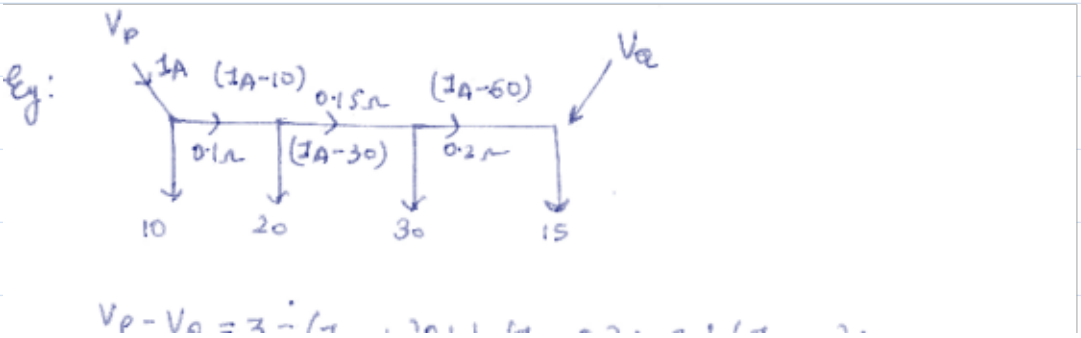
Open e1: feeder currents 8, 3, 1 with loads 5, 2, 1; open e2: 8, 7, 5 with loads 1, 2, 5; open e3: 7, 2, 1 with loads 5, 2, 1

$$P_{L1} = (8^2 + 3^2 + 1^2)r = 74\ r\ ;\ P_{L2} = (8^2 + 7^2 + 5^2)r = 138\ r\ ;\ P_{L3} = (7^2 + 2^2 + 1^2)r = 54\ r$$



Closed loop fed through a source with tapping current IA

Eg:

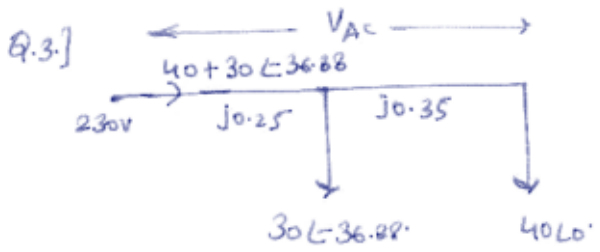


Distributor fed from P (IA) and Q: sections 0.1, 0.15, 0.2 ohm; section currents (IA-10), (IA-30), (IA-60); loads 10, 20, 30, 15 A

$$V_P - V_Q = 3 = (I_A - 10)0.1 + (I_A - 30)0.15 + (I_A - 60)0.2 \rightarrow I_A = 45.55\ A$$

$$V_P = 220 + 15.55 \times 0.15 + (35.55)0.1 = 225.68\ V$$

Q.3]



230 V source, section j0.25 and j0.35 ohm; loads 30 angle -36.88 and 40 angle 0; first section carries 40 + 30 angle -36.88

$$V_{AC} = (30\angle -36.88 + 40)(j0.25) + 40(j0.35)$$